Deep Learning for Image Analysis - Learning from fewer annotations

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2 Strategies

3 Self-supervised Learning (SSL)

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Deep Learning requires massive datasets

- Deep Learning is today the most powerful method for image classification, segmentation and object detecction.
- Deep Learning can achieve or even go beyond human performance for these tasks.
- Problem: deep learning usually relies on massive annotated datasets.
- Example: ImageNet contains 1.4 million annotated images [Russakovsky et al., 2015].

Deep Learning requires massive datasets

- Manual image annotation is annoying.
- Manual image annotation is expensive.
- Sometimes it is even impossible to get a lot of annotated data (for instance rare diseases, security relevant tasks)
- We need to address the need for massive image data sets by:
  - using cheap annotations.
  - using different (abundant) annotated data.
  - using unlabeled data.

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Massive image annotation by crowd sourcing

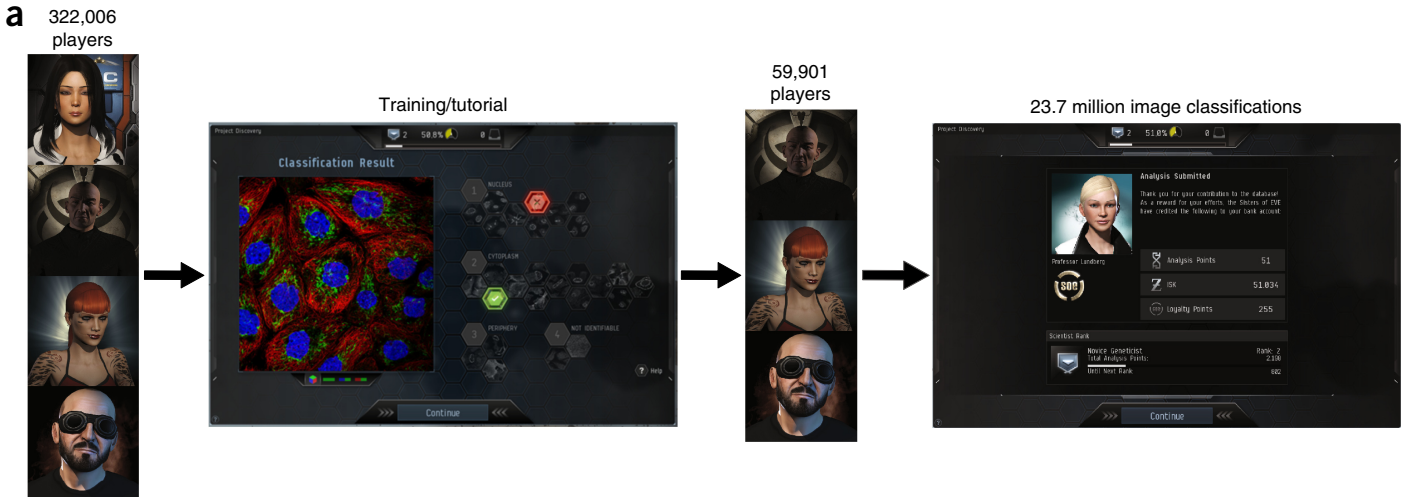


Figure: Gamification for the annotation of protein localization patterns: the actual annotation task is "disguised" as a computer game. Image taken from [Sullivan et al., 2018]

- Crowd sourcing: generate massive annotated data sets by recruiting (potentially untrained) annotators (citizen science).
- Several strategies to reach many people, e.g. gamification.
- Problems:
  - Label noise, heterogeneity of annotation quality
  - Potentially also expensive and laborious

Massive annotation by leveraging routinely acquired data

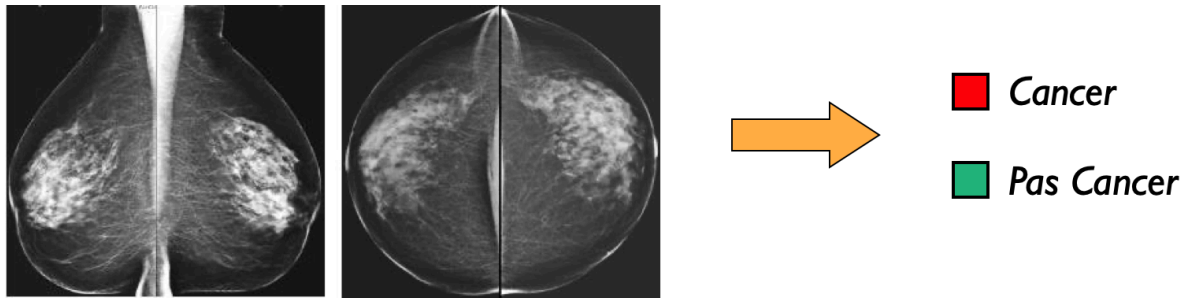


Figure: Radiology: mammographies are acquired routinely by physicians.

- In many fields, image data is routinely acquired (e.g. medical examinations).
- Origins of the image labels:
  - Routine annotation by a medical doctor.
  - Future evolution of the disease.
- Problem: we are limited in the tasks, i.e. in the variables that we can predict.

Massive image annotation by experimental design

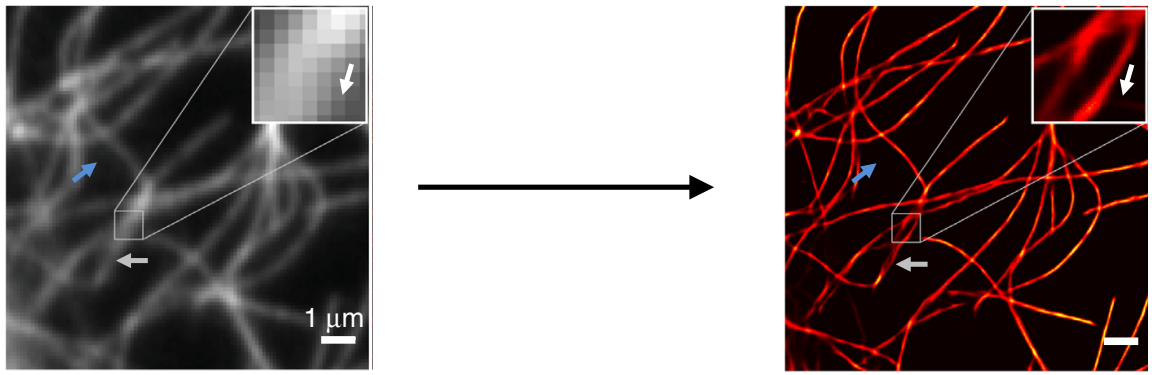


Figure: A high resolution image can be predicted from a low resolution image. Image adapted from [Ouyang et al., 2018]

- In image restoration, we aim at predicting a high quality image from a lower quality image.
- Here, the ground-truth is entirely generated by the experimental design (pairs of images are acquired).
- Similarly, we can also predict different modalities.

Learning from simulated data



Autonomous flying

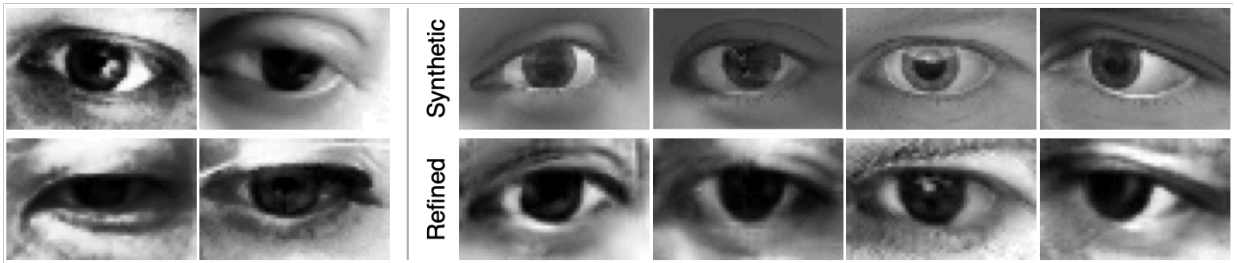


Figure: Two examples for learning from simulated data. Images from [Kang et al., 2019] and [Shrivastava et al., 2017].

- Simulation of large amounts of data with known ground truth and train networks on simulated data.
- Examples: image restoration (artificial degradation of images), scene simulators (autonomous driving).
- Problem: the data distributions between simulated and real data usually differ.

Transfer Learning: learning from other datasets / other tasks

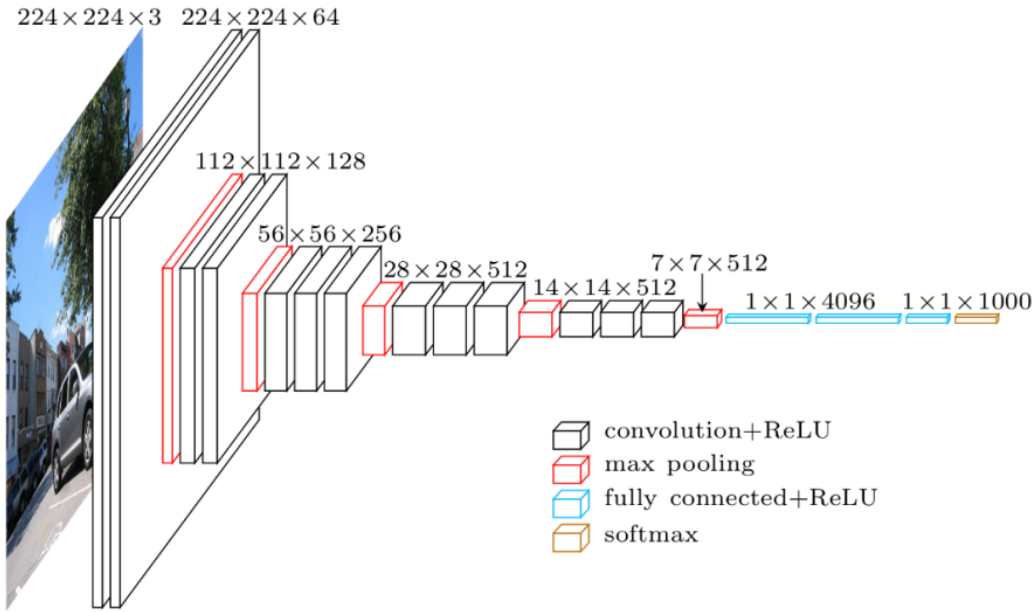


Figure: Image representations for transfer learning.

- Train a network on large annotated image data bases.
- Use the learned representations to solve small scale problems (with or without fine-tuning) with few annotations.
- Problems:
  - Relevance of the used dataset.
  - Relevance of the tasks.

Subject of today

In this lecture, we will learn about four strategies to reduce the need for massive annotated datasets:

1. **Self-supervised learning:** learn representations from unlabeled data.
2. **Weak supervision:** labels exist, but are coarser than the task.
3. **Domain adaptation:** labels exist, but in another domain (e.g. simulated data, other imaging conditions).

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Self-supervised Learning (SSL): General idea

- The success of Deep Learning in Computer Vision stems from the learning of good representations of the image data.
- Opportunity: to use unlabeled data for representation learning.
- Problem: data without a task is not usable to train Neural Networks.
- Idea: define a pretext task that is easily available without annotation.
- Self Supervision: supervised learning on labels that are directly and automatically generated from the data itself.

Early approach: training on spatial context of image patches

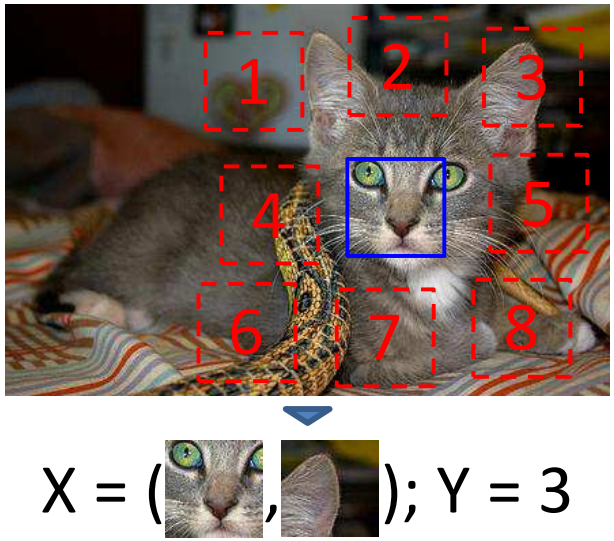


Figure: Self-supervised learning using spatial context as a pretext task[Doersch et al., 2015]

- One patch x_q is randomly sampled from the 8 positions around the central patch x_c.
- x_c and x_q are encoded (same network), and the encodings are fused.
- The prediction task is the position of x_q with respect to x_c.

Early approach: autoencoders and their variants

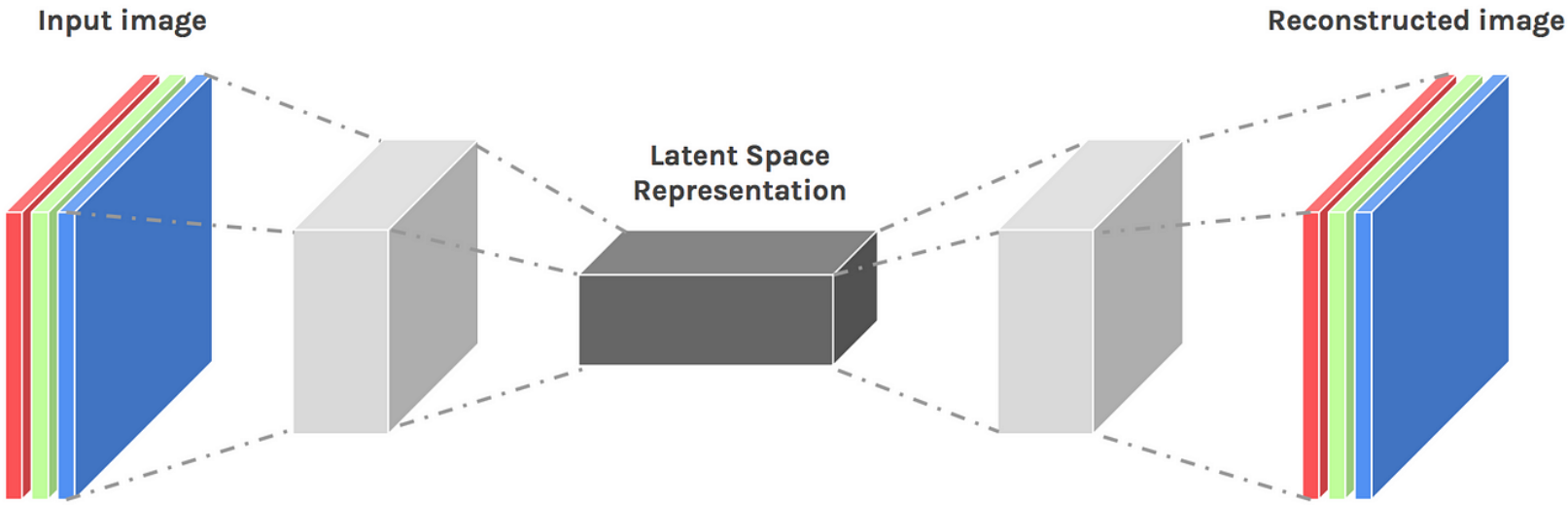


Figure: Autoencoder: the pretext task is identical reconstruction.

- Pretext task: reconstruction of the image
- Many variants
  - denoising autoencoders: input is corrupted by noise [Vincent et al., 2010].
  - variational autoencoders: generative models based on autoencoders
- Autoencoders will be treated in a later lecture.

Contrastive Learning / Siamese representation learning

- Pretext task:
  - For each image, we generate *views* (i.e. transformed versions of the original image)
  - We train a network to maximize agreement between representations of views originating from the same image.
- The representations are thus supposed to be robust with respect to the transformations.
- One problem is the "collapse": a situation where all representations are 0 (trivial solution).
- We will see a series of famous papers that have proposed different solutions to this problem.

A simple framework for Contrastive Learning of Visual Representations (SimCLR) [Chen et al., 2020]

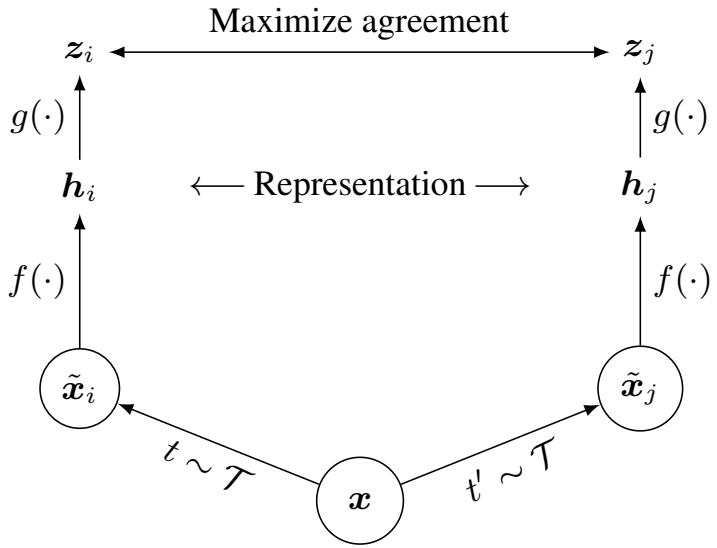


Figure: SimCLR workflow as defined in [Chen et al., 2020]

- For each image x, we calculate two transformed versions $\tilde{x}_i$ and $\tilde{x}_j$ by two transformations t and t' drawn from a set of parametrized transformations.
- For $\tilde{x}_i$ and $\tilde{x}_j$, we calculate the representations h_i and h_j, respectively. $f(\cdot)$ is the neural network we want to learn.

SimCLR [Chen et al., 2020]

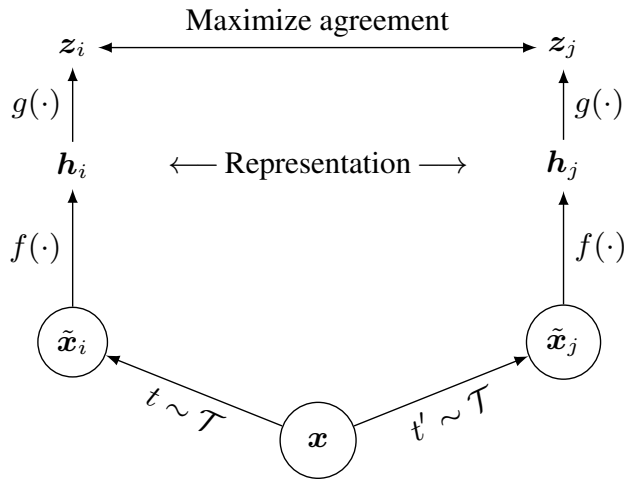


Figure: SimCLR workflow as defined in [Chen et al., 2020]

- The representations h_i and h_j are mapped to a space where contrastive loss is applied, by a small neural network $g(\cdot)$.
- The task is thus to maximize the agreement between z_i and z_j, i.e. we wish z_i and z_j to be close.

SimCLR [Chen et al., 2020]

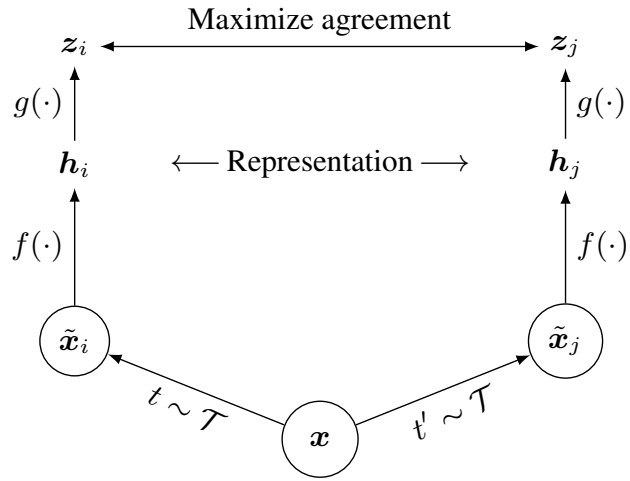


Figure: SimCLR workflow as defined in [Chen et al., 2020]

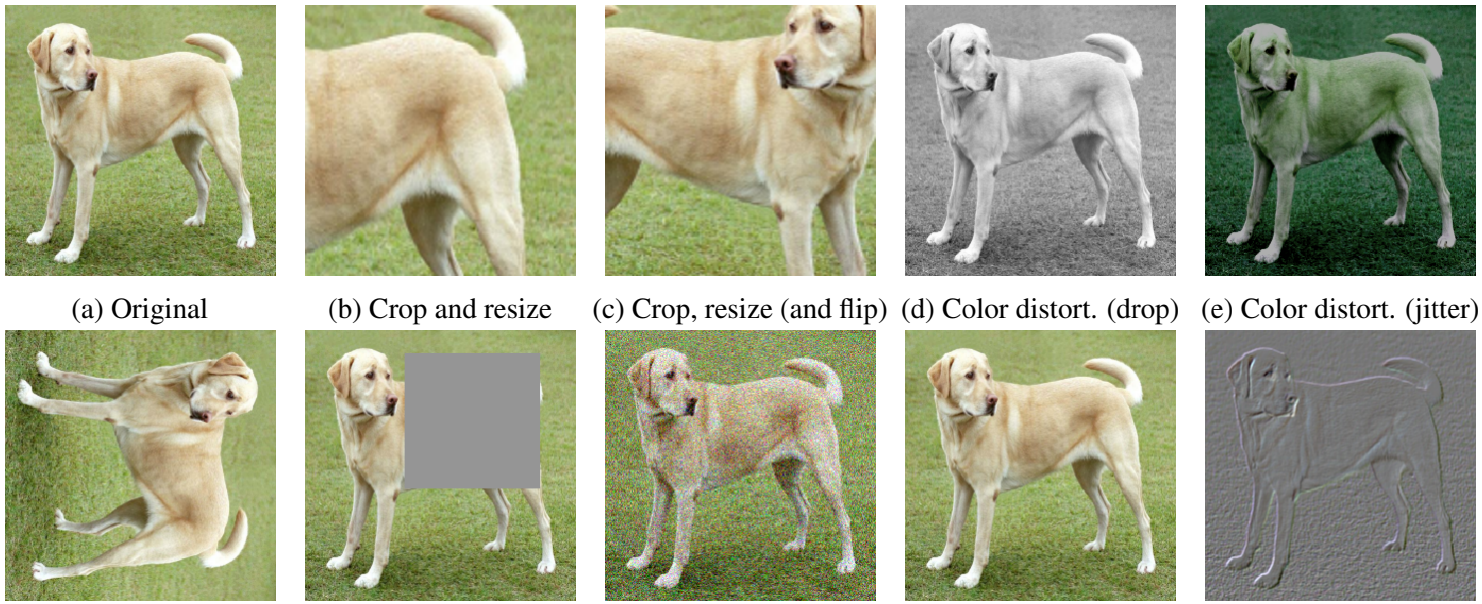
- We define the similarity of two representations as follows:

$$s(u, v) = \frac{u^T v}{\|u\|\|v\|} \tag{1}$$

- With this, we can define the contrastive loss as follows:

$$l_{i,j} = -\log \frac{\exp \frac{s(z_i, z_j)}{\tau}}{\sum_{k=1}^{2N} \mathbb{1}_{k \neq i} \exp \frac{s(z_i, z_k)}{\tau}} \tag{2}$$

SimCLR: transformations



(a) Original (b) Crop and resize (c) Crop, resize (and flip) (d) Color distort. (drop) (e) Color distort. (jitter)

(f) Rotate $\{90°, 180°, 270°\}$ (g) Cutout (h) Gaussian noise (i) Gaussian blur (j) Sobel filtering

Figure: Transformations used in SimCLR [Chen et al., 2020]

SimCLR: results

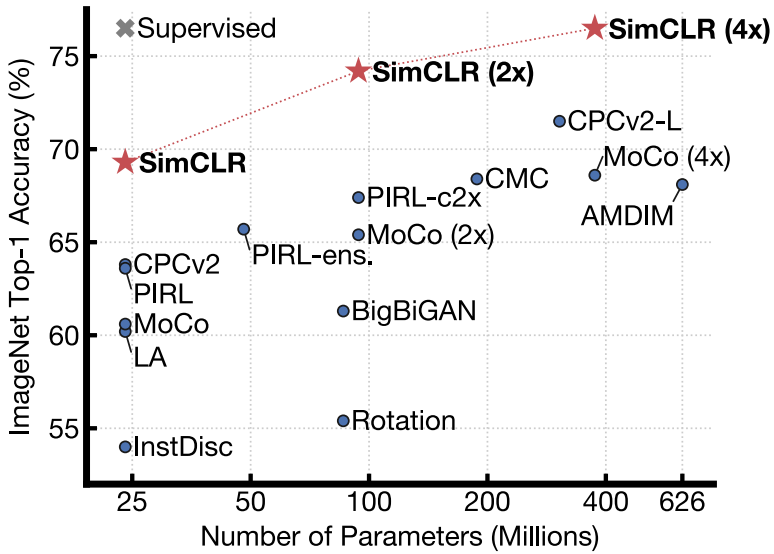


Figure: SimCLR: Results (linear evaluation) [Chen et al., 2020]

- Evaluation of representations by *linear evaluation* (result obtained by a linear classifier trained with the representation as features): the unsupervised approach comes reasonably close to a fully supervised approach.

SimCLR: learning with few labels

- SimCLR provides powerful and generic representations.
- Fine-tuning the pre-trained network on only 10% of the labels comes close to a network trained on 100% of the labels.
- Take-home message: unlabeled data reduces the number of labels we need.

MOCO

- Momentum Contrast for Unsupervised Visual Representation Learning[He et al., 2020]
- Contrastive Learning approach (same idea as SimCLR):
  - generate two views of an image.
  - train a network to get encodings, where these two views are close, and those of other images are far.
- Challenge: in contrastive Learning, we need large batch sizes (many negative samples, i.e. many keys):
  - Makes the problem harder, and the harder the problem the better the encodings.
  - Negative samples can be virtually anything. So we need sample this large visual space.
- In the following, we compare the query image representation q with the key image representations k (one of which comes from the same image that gave rise to q).

MOCO

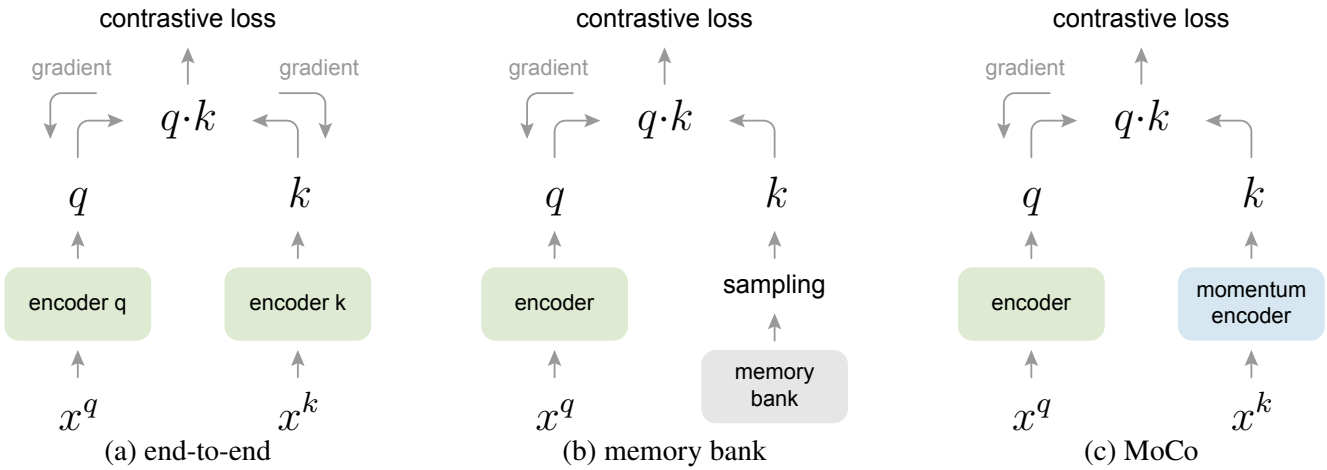


Figure: Different settings for contrastive learning [He et al., 2020]

1. End-to-end: for each batch, all keys and queries are encoded on the fly, and the encoder is updated both for negative and for positive samples.
2. Memory Bank: the representations of negative samples are not calculated online but taken from a memory bank. The encoder is updated only for the query image.
3. MoCo: maintaining a queue of encoded data samples, which are successively replaced by new samples, encoded with the new version of the key encoder.

MOCO: Strategy

- The queue of keys contains (many) encoded samples.
- When a mini-batch is processed:
  - two views of the each image are generated and processed (the query view and the positive key view) with the current version of the encoder.
  - After processing, the oldest encoded samples are removed, and the new key samples are added to the queue, serving as negatives for the next comparisons.
  - The momentum encoder is updated with the updated query encoder:

$$\theta_k \leftarrow m\theta_k + (1 - m)\theta_q \tag{3}$$

- For training the contrastive task, we thus have access to a large number of negative samples, however not encoded identically.

MOCO results

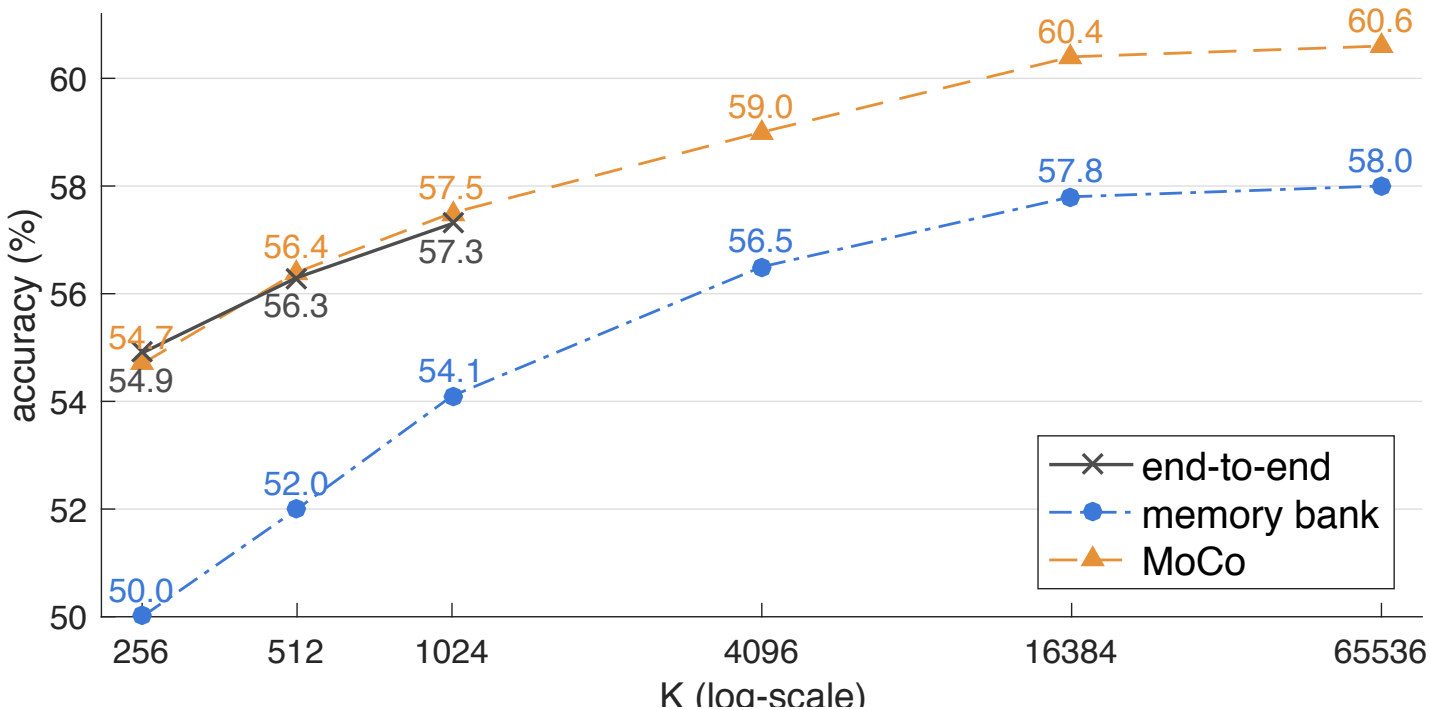


Figure: Results depending on the number of negative samples [He et al., 2020]

BYOL (2020): Bootstrap Your Own Latent

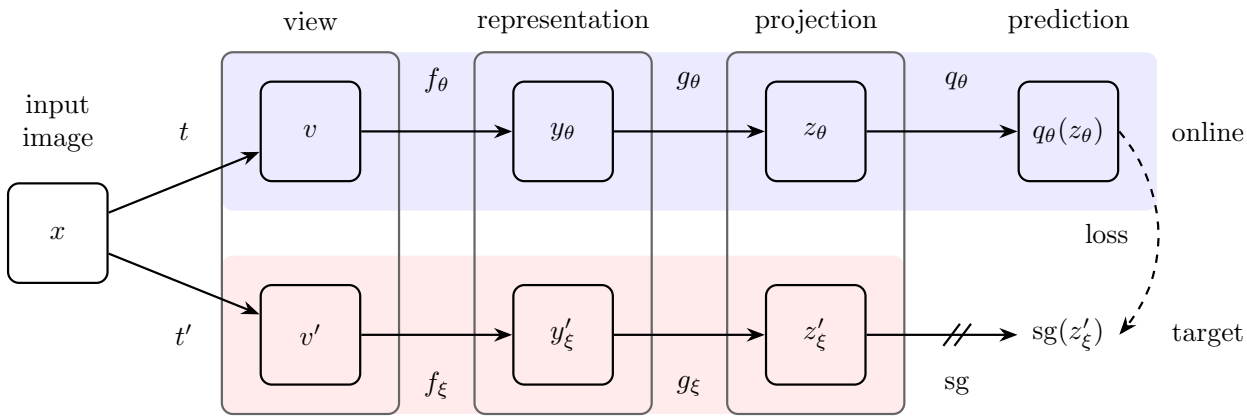


Figure: BYOL principle [Grill et al., 2020]

- Again, we generate two views for the same input image.
- The two views are input of two different branches: the online branch and the target branch.
- The design is deliberately asymmetric, between the online and the target branch.

BYOL (2020): Bootstrap Your Own Latent

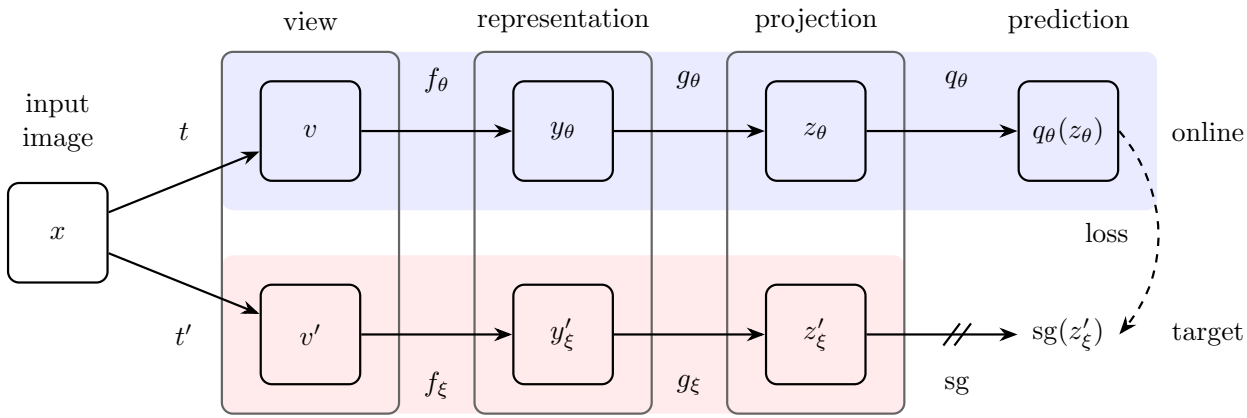


Figure: BYOL principle [Grill et al., 2020]

- **Online**: $y = q_\theta(g_\theta(f_\theta(v)))$ (has an additional predictor q).
- **Target**: $y' = g_\xi(f_\xi(v'))$ (no predictor).
- The parameters of the target branch are not optimized, but updated by an exponentially moving average : $\xi \leftarrow \tau\xi + (1-\tau)\theta$.

BYOL (2020): Bootstrap Your Own Latent

- **Loss**:

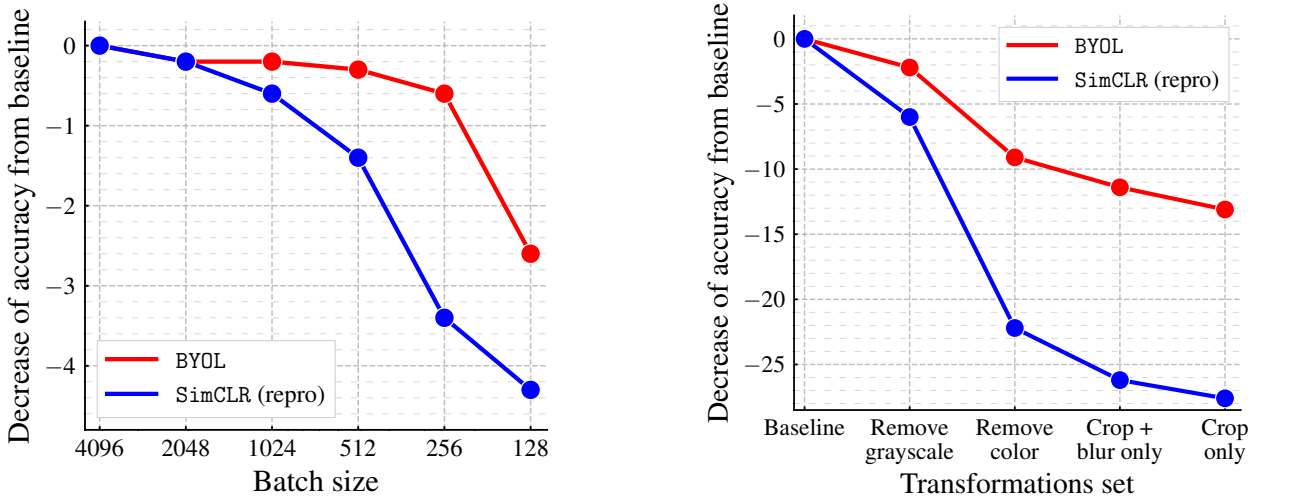
$$\mathcal{L}_{\theta,\xi} = \|\bar{q}_\theta(z_\theta) - \bar{z}'_\xi\|_2^2 = 2 - 2 \cdot \frac{\left\langle q_\theta(z_\theta), z'_\xi \right\rangle}{\|q_\theta(z_\theta)\|_2 \|z'_\xi\|_2} \tag{4}$$

- $\bar{\cdot}$ denotes the ℓ_2-normalized vector: $\bar{a} = a/\|a\|_2$.
- No explicit negatives, no contrastive denominator.
- The representation is trained to "chase" a slowly moving target.

BYOL: why does it not collapse?

- A trivial solution exists: all images mapped to the same vector gives zero loss.
- BYOL avoids it empirically by three ingredients:
  - **Stop-gradient:** the loss is only backpropagated through the online network; the target is treated as a constant.
  - **Slowly moving target:** the target network is an EMA of the online network.
  - **Asymmetry:** the predictor q_θ exists only on the online branch.

BYOL (2020): Bootstrap Your Own Latent



(a) Impact of batch size

(b) Impact of progressively removing transformations

Figure: Comparison between simCLR and BYOL [Grill et al., 2020]

- Less dependency on batch size, as no negative examples are needed.
- Less dependency on the transformations: the task never becomes simple, as it can happen in simCLR

DINO (2021): Self-distillation with no labels

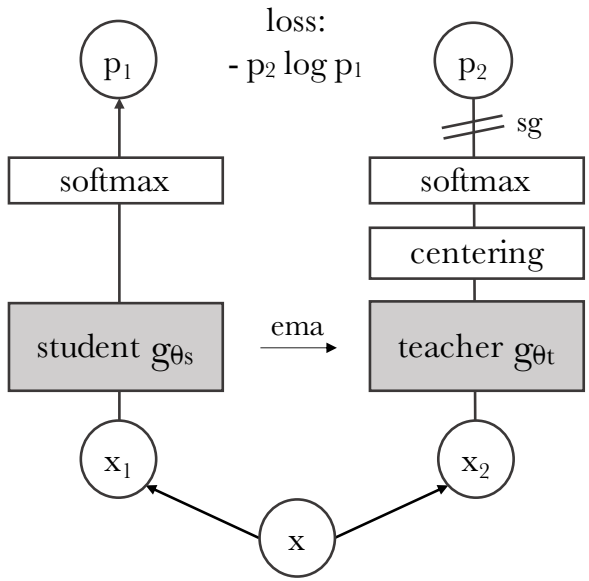


Figure: DINO principle [Caron et al., 2021]

Similar architecture and same principle as BYOL[Grill et al., 2020]:

- Teacher-student architecture: the teacher provides the target for the student
- The teacher network is updated by exponentially moving average (EMA) from the student network.
- No negatives (no contrastive learning).

Principle 1/2

- Two networks with shared architecture: Student g_{θ_s} and Teacher g_{θ_t} (EMA of student).
- Like in contrastive learning, multiple views of each image are generated:
  - 2 *global* views x_1^g, x_2^g (same resolution as x, here 224×224).
  - several *local* views of smaller resolution (here: 96×96).
- Teacher and student generate output probability distributions over K dimensions:

$$P_s(x) = \mathrm{softmax}\left(\frac{h_{\theta_s}(f_{\theta_s}(x))}{\tau_s}\right), \quad P_t(x) = \mathrm{softmax}\left(\frac{h_{\theta_t}(f_{\theta_t}(x))}{\tau_t}\right)$$

- Here $f_\theta(\cdot)$ is the feature extractor (or backbone), and $h_\theta(\cdot)$ is the projection head. For downstream tasks, we only use f_θ. The complete networks are $g = h \circ f$ for teacher and student respectively.

Principle 2/2

- During training, the student aims at approaching the teacher distributions P_t.
- Training (V is the full set of views):

$$\min_{\theta_s} \sum_{x \in \{x_1^g, x_2^g\}} \sum_{\substack{x' \in V \\ x \neq x'}} -P_t(x) \log P_s(x') \tag{5}$$

- Teacher parameters are not updated by backpropagation, but via EMA:

$$\theta_t \leftarrow \lambda \theta_t + (1 - \lambda)\theta_s \tag{6}$$

DINO: avoiding collapse by centering and sharpening

- Two possible collapses of the teacher output P_t:
  - one dimension dominates for all images,
  - uniform distribution for all images.
- **Centering** (against domination): subtract a running mean c from the teacher output:

$$P_t(x) = \mathrm{softmax}\left(\frac{g_{\theta_t}(x) - c}{\tau_t}\right)$$

$$c \leftarrow m\,c + (1-m)\frac{1}{B}\sum_{i=1}^{B} g_{\theta_t}(x_i)$$

- **Sharpening** (against uniformity): low teacher temperature $\tau_t < \tau_s$.
- The two effects balance each other; together with the EMA teacher, no negatives are needed.
- Teacher sees only the *global* views, the student sees all views: "local-to-global" correspondence.

DINO: emergent properties with Vision Transformers

- With ViT backbones, the attention maps of the [CLS] token segment the main object, although no segmentation label was ever used.
- Features work very well with a simple k-NN classifier: semantically similar images are close in the representation space.
- This made DINO the basis of many later foundation models.

Masked image modeling: MAE [He et al., 2022]

- Idea from NLP (BERT): hide part of the input and predict it.
- The image is split into patches; a large fraction (75%) is **masked** at random.
- The ViT encoder sees only the visible patches; a small decoder reconstructs the masked ones.
- Loss on the masked patches M only:

$$\mathcal{L} = \frac{1}{|M|} \sum_{i \in M} \|x_i - \hat{x}_i\|_2^2$$

- High masking ratio makes the task hard (no trivial interpolation) and training cheap (encoder processes 25% of patches).
- Unlike contrastive methods: no augmentations needed for the pretext task, no collapse problem.

DINOv2: scaling self-supervision [Oquab et al., 2024]

- Combines two losses:
  - image-level: DINO loss on the [CLS] token,
  - patch-level: masked patch prediction in feature space (iBOT [Zhou et al., 2022]).
- Trained on a **curated** dataset of 142M images: data quality matters as much as data quantity.
- Large ViT (1B parameters), distilled into smaller models.
- Frozen features (no fine-tuning) are competitive with supervised models for classification, segmentation and depth estimation.
- DINOv3 [Siméoni et al., 2025] scales this further (7B parameters) and improves dense (patch-level) features.

Foundation models

- **Foundation model** [Bommasani et al., 2021]: a large model trained on broad data, mostly without labels, that can be adapted to many downstream tasks.
- Adaptation with few or no labels: linear probing, fine-tuning, or zero-shot (e.g. with text prompts).
- Examples in image analysis:
  - **General vision:** DINOv2 [Oquab et al., 2024] (generic image features), CLIP [Radford et al., 2021] (images and text, zero-shot classification), SAM [Kirillov et al., 2023] (promptable segmentation).
  - **Remote sensing:** SatMAE [Cong et al., 2022] (MAE on multi-spectral satellite images).
  - **Computational pathology:** UNI [Chen et al., 2024], Virchow [Vorontsov et al., 2024] (DINOv2 on millions of histology tiles).
- Common idea: pay the cost of pretraining once, then solve many tasks with few labels.

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What is weak supervision?

- Weak supervision refers to a situation, where the ground truth data we build our model on is in some sense imperfect [Zhou, 2018].
- Three types of weak supervision:
  1. **Incomplete supervision:** only a subset of training data are labeled (semisupervised setting).
  2. **Inexact supervision:** training data are labeled but not as exactly as required by the task we would like to perform.
  3. **Inaccurate supervision:** training data are labeled, but may contain mistakes.
- Here, we treat the case of inexact supervision.

Multiple Instance Learning (MIL): example



Figure: Principle of Multiple Instance Learning. Image taken from [Babenko, 2008]

- Key-example: we are given a number of persons each of which has several keys. You are just given the information who is able to open the door.
- Examples:
  - Video tags without frame annotation
  - Medical imaging: image is positive in the presence of a lesion
  - Marketing: campaign sent out to a group of people

Multiple Instance Learning (MIL)

- Instead of labels y_j for each instance x_j, we only have labels for **bags** of instances:

$$T = \{(B_i, Y_i)\}_{i=1}^{N}, \qquad B_i = \{x_{i1}, \ldots, x_{iK_i}\}$$

- Standard MIL assumption: a bag is positive if *at least one* instance is positive, negative if all instances are negative:

$$Y_i = \max_k y_{ik}$$

- The number of instances K_i can vary from bag to bag.
- The model must be **permutation invariant**: the order of instances does not matter.

Application: whole slide images (WSI) in pathology

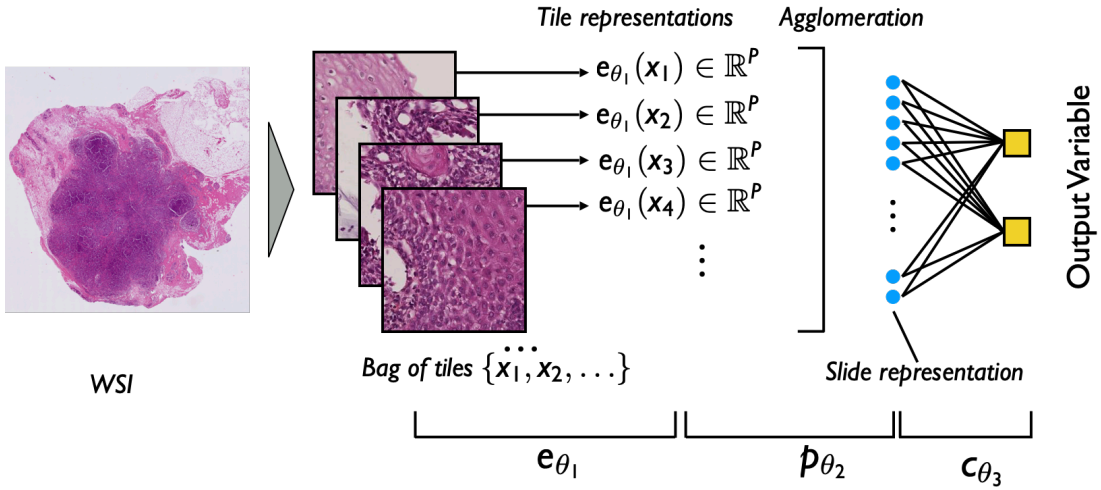


Figure: Classification of whole slide images.

- A WSI can be $100{,}000 \times 100{,}000$ pixels: it does not fit into a GPU.
- Labels are available at the slide or patient level (diagnosis, mutation, survival), rarely at the region level.
- The relevant region can be very small (e.g. a micro-metastasis).
- MIL: **slide = bag**, **tiles = instances**.

Application: whole slide images (WSI) in pathology

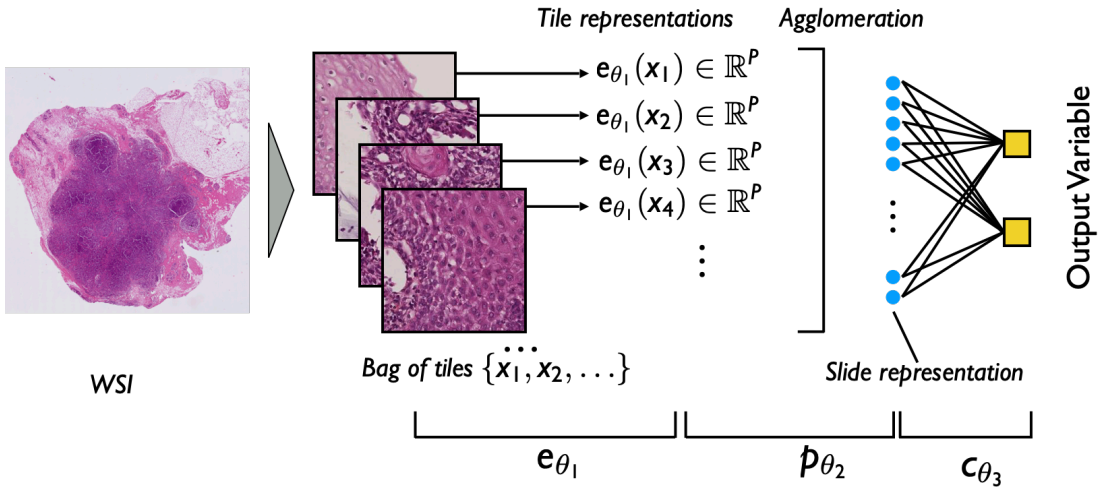


Figure: Classification of whole slide images.

1. Tissue detection and tiling (e.g. 224×224 pixels at $20\times$): thousands of tiles per slide.
2. Feature extraction with a pretrained encoder (today: SSL foundation model), computed once.
3. A small **aggregation** and downstream network, trained with slide labels only.

Only the aggregation and downstream parts are trained: feasible with a few hundred slides.

Aggregation functions

Given tile embeddings $z_1, \ldots, z_K$, the slide representation a can be:

- **Max pooling:** exactly the MIL assumption (one positive tile is enough) [Campanella et al., 2019].
  Works for small lesions, but only one tile receives a gradient: unstable, ignores context.
- **Mean pooling:** $a = \frac{1}{K} \sum_k z_k$.
  Stable, but small relevant regions are diluted.
- **Attention pooling:** $a = \sum_k s_k z_k$, with weights s_k *learned* from the tiles [Ilse et al., 2018].
  Compromise between max and mean: the model learns which tiles matter.

All three are permutation invariant and handle a variable number of tiles.

Example: Metastasis detection in lymph nodes

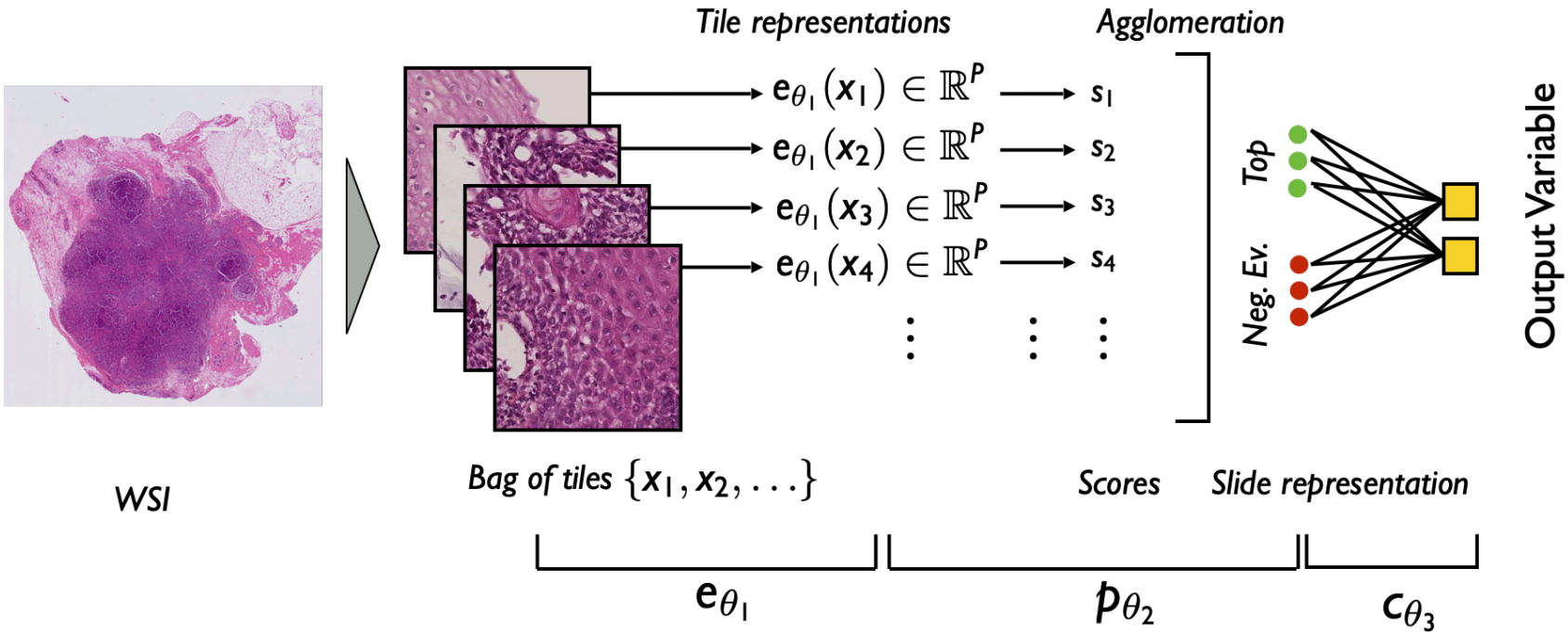


Figure: Attention-based approach for metastasis detection.

- A slide is metastatic if one tile is metastatic.
- Max-Pooling or variants are appropriate

Application example: metastasis detection

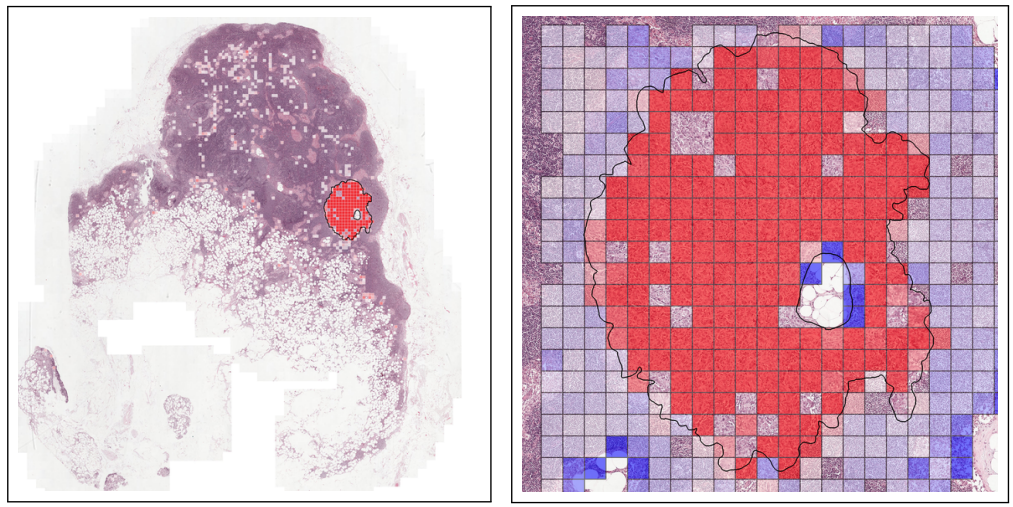


Figure: Example for MIL for tumor detection. Image adapted from [Courtiol et al., 2017]

Here, [Courtiol et al., 2017] use K top and K bottom scores to detect metastases. Only global annotation is used. The results are on par with classification / detection algorithms trained on detailed annotation.

Attention-based MIL [Ilse et al., 2018]

- Each tile embedding z_k gets a score, normalized over the bag by a softmax:

$$s_k = \frac{\exp\left(w^\top \tanh(Vz_k)\right)}{\sum_{j=1}^{K} \exp\left(w^\top \tanh(Vz_j)\right)}$$

  where V and w are learned parameters.
- Slide representation and prediction:

$$a = \sum_{k=1}^{K} s_k z_k, \qquad \hat{Y} = \sigma(c^\top a)$$

- Trained end-to-end with a standard classification loss on slide labels.

Attention maps: localization for free?

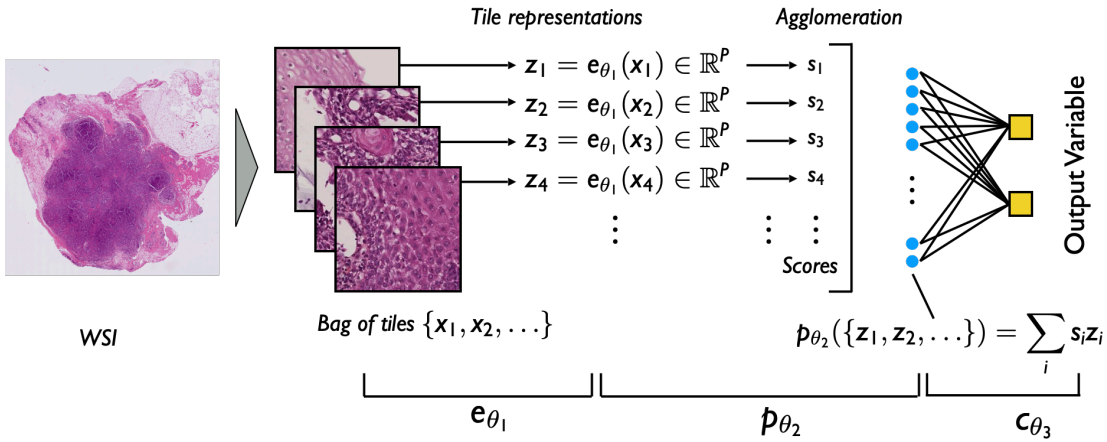


Figure: Attention-based approach for histology image classification.

- Projecting s_k back onto the slide gives a heatmap of the regions used for the decision.
- Useful for interpretation and for checking that the model looks at tissue, not artifacts.
- But: attention ≠ explanation. High attention does not mean the tile is positive; it means the tile was important for the decision (positive or negative evidence).

Weak supervision in pathology: discussion

- Slide-level labels come for free from clinical routine: diagnosis, molecular status, outcome.
- This allows predicting variables that pathologists cannot see by eye (e.g. mutations, response to treatment).
- Current pipelines combine the strategies of this lecture:
  - **self-supervision** to train the tile encoder (no labels),
  - **weak supervision** to train the aggregation (slide labels).
- Limitations: needs enough slides (hundreds), sensitive to confounders (center, scanner, staining): the model may learn the hospital instead of the biology $\rightarrow$ domain adaptation.

Overview

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Example: Simulation for the generation of training data

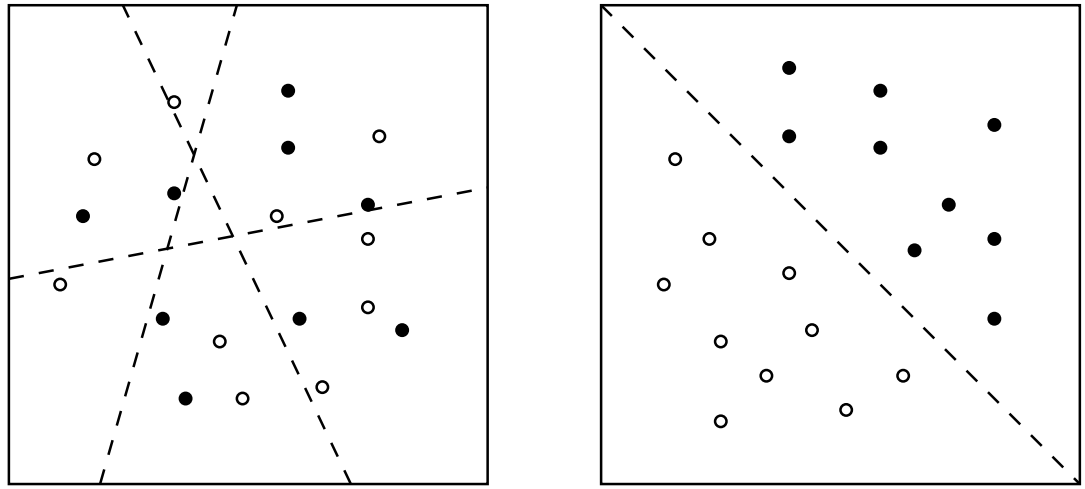
- With simulation, we can generate arbitrary quantities of data with known ground truth.
- Simulation is regularly used for benchmarking of methods.
- Usually, simulation provides us with data similar to real world data, but not identical.
- For this reason, the use of simulated data for training neural networks is limited.
- Idea: can we overcome the differences between simulated and real data by explicit algorithmic strategies?

Domains

- Domain adaptation: learning a discriminative classifier when training data does not follow the same distribution as the test data.
- Let $\mathcal{X}$ be an input space and $\mathcal{Y} = \{1, \ldots, L\}$ the set of L possible labels. We call a domain the distribution over $\mathcal{X} \times \mathcal{Y}$.
- Here we consider two domains: the source domain $\mathcal{D}_S$ and the target domain $\mathcal{D}_T$.
- The n source samples are then drawn from $\mathcal{D}_S$, the $N - n$ test samples from $\mathcal{D}_T$:

$$\begin{aligned} S &= \{(x_i, y_i)\}_{i=1}^{n} \sim \mathcal{D}_S^n \\ T &= \{(x_i, y_i)\}_{i=n+1}^{N} \sim \mathcal{D}_T^{N-n} \end{aligned}$$

The divergence of domains



(a) Low divergence (b) High divergence

Figure: Illustration of h-divergence between domains.

- The divergence of domains can be quantified by trying to classify samples according to their source label.
- Hence, we train a classifier from a training set $T = \{(x_i, d_i)\}$:

$$d_i = \begin{cases} 0, & \text{if } x_i \sim \mathcal{D}_S^X \\ 1, & \text{if } x_i \sim \mathcal{D}_T^X \end{cases} \tag{7}$$

- In this case, the h-divergence can be written as $2(1 - 2\epsilon)$, where ϵ is the error of the classifier.

Domain adaptation by adversarial training

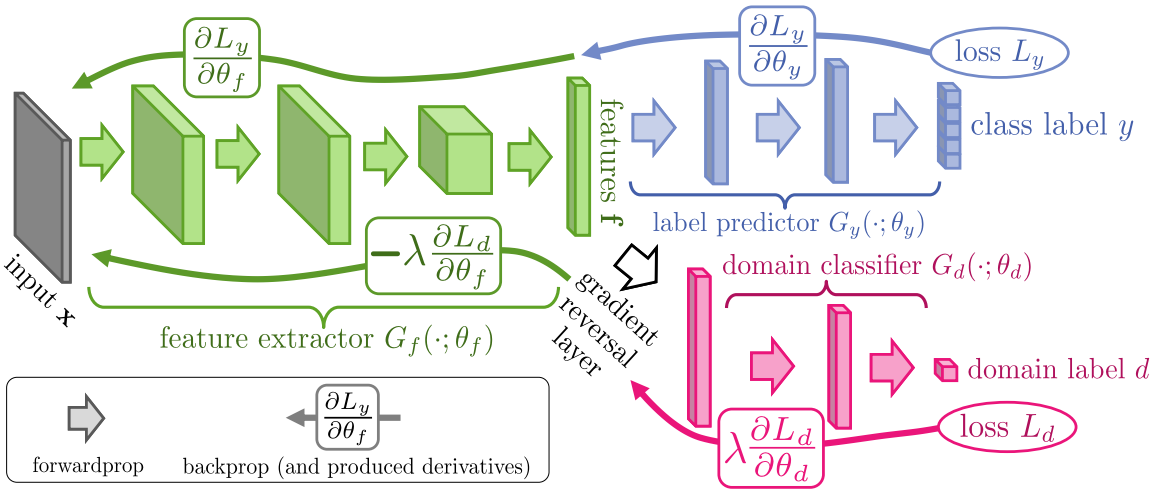


Figure: Domain adaptation by adversarial training. Image taken from [Ganin et al., 2016]

- We seek a representation that provides good prediction results, but low h-divergence.
- This means that this representation (f in the figure) should not allow us to distinguish between source domain and target domain.

Domain adaptation by adversarial training

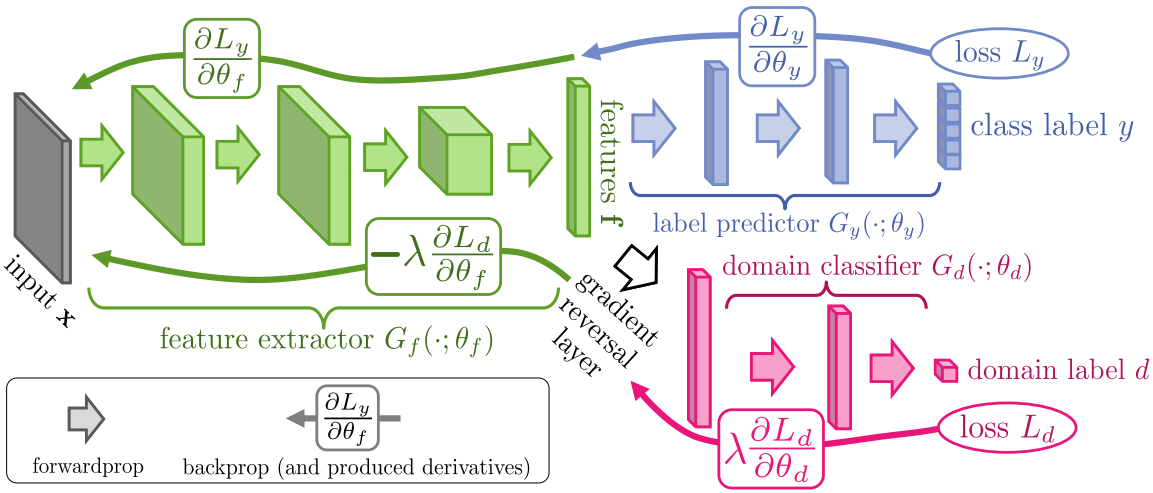


Figure: Domain adaptation by adversarial training. Image taken from [Ganin et al., 2016]

For this we consider the following architecture [Ganin et al., 2016]:

- $G_f(;,\theta_f)$: feature extractor
- $G_y(;,\theta_y)$: label predictor
- $G_d(;,\theta_d)$: domain classifier

Domain adaptation by adversarial training

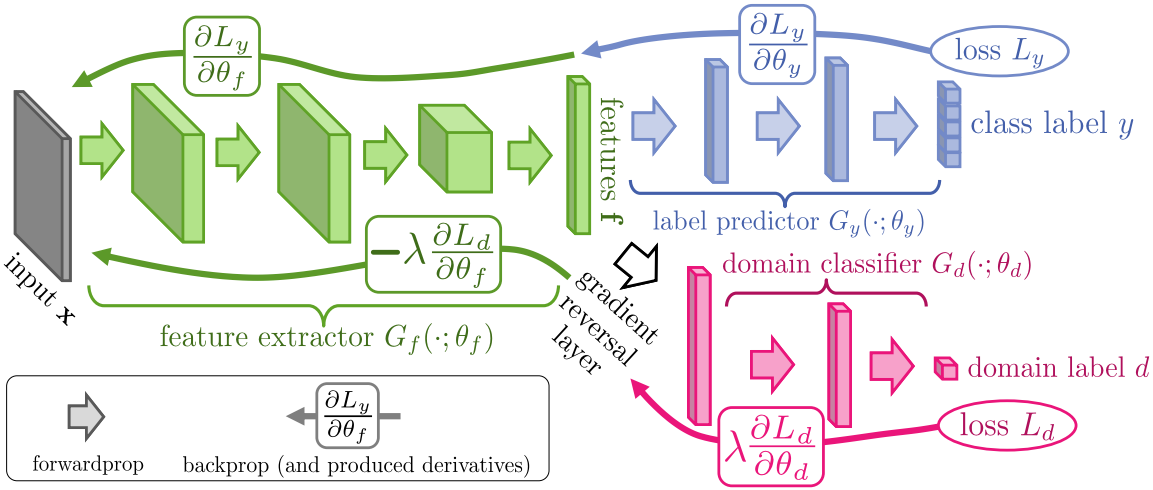


Figure: Domain adaptation by adversarial training. Image taken from [Ganin et al., 2016]

We consider the following loss:

$$L(\theta_f, \theta_y, \theta_d) = L_y(\theta_f, \theta_y) - \lambda L_d(\theta_f, \theta_d) \tag{8}$$

where $L_y(\theta_f, \theta_y)$ is the standard prediction loss for samples from the training set (and thus the source domain), while $L_d(\theta_f, \theta_d)$ is the domain loss, calculated on all samples (labeled samples from the source domain and the unlabeled samples from the target domain).

Domain adaptation by adversarial training - Optimization

- More formally, we write:

$$
\begin{aligned}
L_y(\theta_f, \theta_y) &= \frac{1}{n} \sum_{i=1}^{n} L_y(G_y(G_f(x_i, \theta_f), \theta_y), y_i) \\
L_d(\theta_f, \theta_d) &= \frac{1}{n} \sum_{i=1}^{n} L_d(G_d(G_f(x_i, \theta_f), \theta_d), d_i) \\
&+ \frac{1}{N-n} \sum_{i=n+1}^{N} L_d(G_d(G_f(x_i, \theta_f), \theta_d), d_i) \\
L(\theta_f, \theta_y, \theta_d) &= L_y(\theta_f, \theta_y) - \lambda L_d(\theta_f, \theta_d)
\end{aligned}
$$

- This loss has to be minimized with respect to some parameters and maximized with respect to other parameters:

$$
\begin{aligned}
(\hat{\theta}_f, \hat{\theta}_y) &= \arg\min_{\theta_f, \theta_y} L(\theta_f, \theta_y, \hat{\theta}_d) \\
\hat{\theta}_d &= \arg\max_{\theta_d} L(\hat{\theta}_f, \hat{\theta}_y, \theta_d)
\end{aligned}
$$

Domain adaptation by adversarial training - gradient reversal

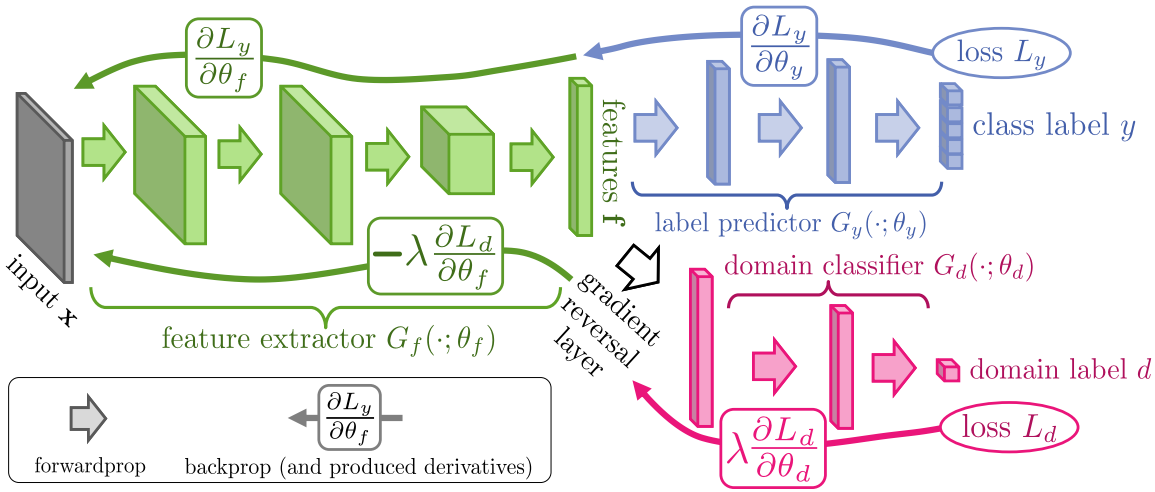


Figure: Domain adaptation by adversarial training. Image taken from [Ganin et al., 2016]

The solution of simultaneous minimization and maximization can be elegantly solved by reversing a gradient layer during back-propagation, as indicated in the figure.

Application example: character classification

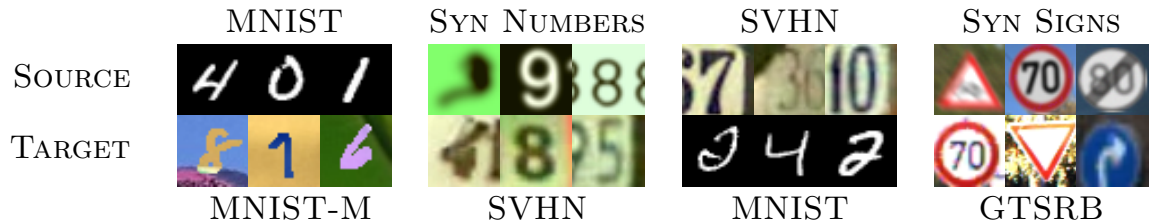
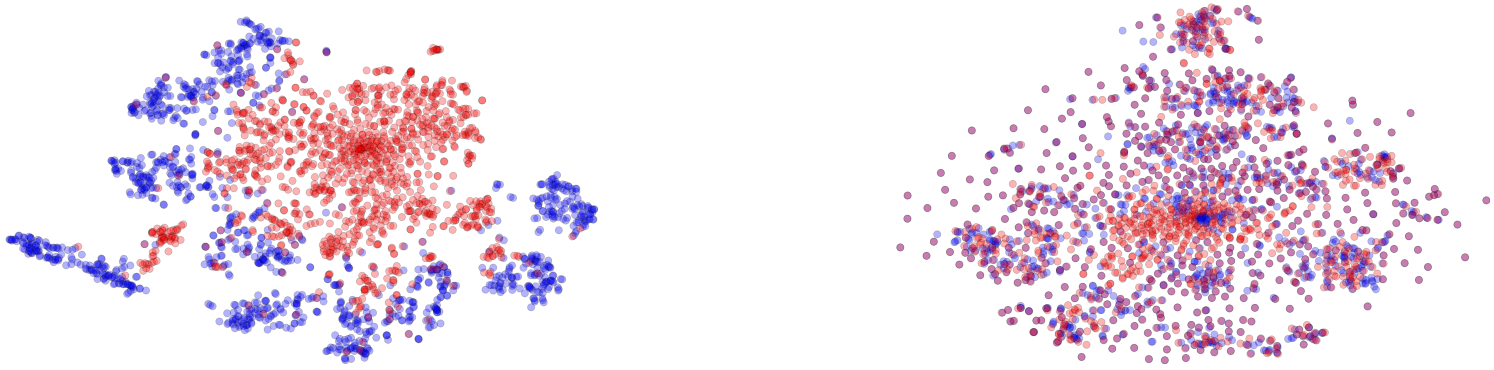


Figure: Domain adaptation between simulated and real data. MNIST: hand-written digits, MNIST-M: MNIST on top of random image patches, SYN Numbers: simulation created by varying Windows fonts, SVHN: street view house numbers, Image taken from [Ganin et al., 2016]

Results indicate that the achievable improvements can reach up to 20% in cases where the simulations are relatively far from the real-world examples.

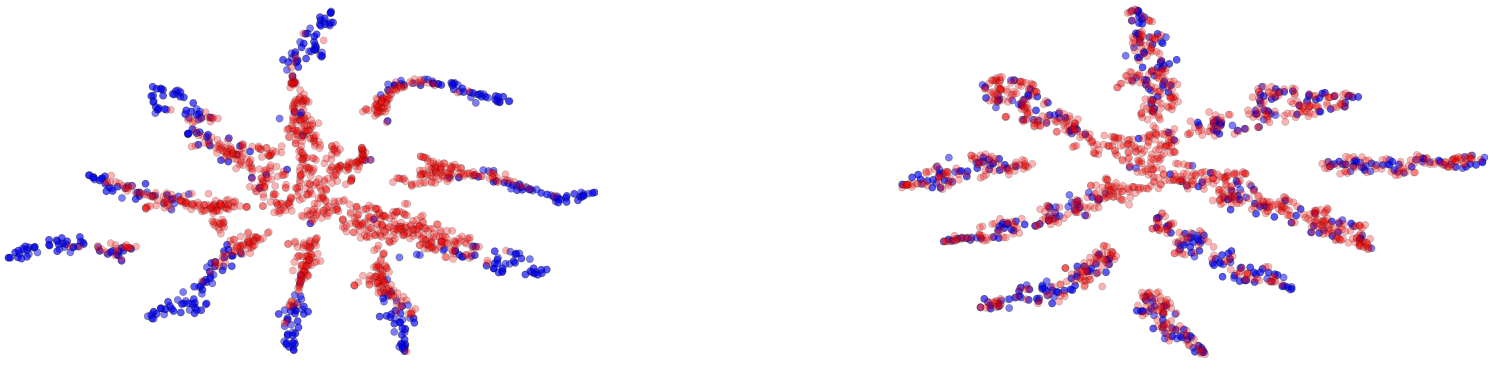
Application example: character classification

MNIST → MNIST-M: top feature extractor layer



(a) Non-adapted (b) Adapted

SYN NUMBERS → SVHN: last hidden layer of the label predictor



(a) Non-adapted (b) Adapted

Figure: Learned representations with and without domain adaptation. Domains are colored in red and blue. Image taken from [Ganin et al., 2016].

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Conclusion

- Large annotated datasets are a bottleneck for many applications, in particular in medicine.
- Four strategies:
  1. **Self-supervised learning:** representations from unlabeled data; basis of **foundation models**.
  2. **Weak supervision:** coarse labels, e.g. MIL for whole slide images.
  3. **Domain adaptation:** transfer from a labeled source domain to an unlabeled target domain.
- In practice, they are combined: e.g. SSL foundation model + attention MIL in pathology.
- **Foundation models** are trained on broad unlabeled data and can be adapted to many tasks with few labels.

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